

Preface

In the 30 years since the publication of the first edition of this book, much has changed in the field of mathematics known as dynamical systems. In the early 1970s, we had very little access to high-speed computers and computer graphics. The word chaos had never been used in a mathematical setting, and most of the interest in the theory of differential equations and dynamical systems was confined to a relatively small group of mathematicians.

Things have changed dramatically in the ensuing 3 decades. Computers are everywhere, and software packages that can be used to approximate solutions of differential equations and view the results graphically are widely available. As a consequence, the analysis of nonlinear systems of differential equations is much more accessible than it once was. The discovery of such complicated dynamical systems as the horseshoe map, homoclinic tangles, and the Lorenz system, and their mathematical analyses, convinced scientists that simple stable motions such as equilibria or periodic solutions were not always the most important behavior of solutions of differential equations. The beauty and relative accessibility of these chaotic phenomena motivated scientists and engineers in many disciplines to look more carefully at the important differential equations in their own fields. In many cases, they found chaotic behavior in these systems as well. Now dynamical systems phenomena appear in virtually every area of science, from the oscillating Belousov-Zhabotinsky reaction in chemistry to the chaotic Chua circuit in electrical engineering, from complicated motions in celestial mechanics to the bifurcations arising in ecological systems.

As a consequence, the audience for a text on differential equations and dynamical systems is considerably larger and more diverse than it was in the 1970s. We have accordingly made several major structural changes to this text, including the following:

1. The treatment of linear algebra has been scaled back. We have dispensed with the generalities involved with abstract vector spaces and normed linear spaces. We no longer include a complete proof of the reduction of all $n \times n$ matrices to canonical form. Rather we deal primarily with matrices no larger than 4×4 .
2. We have included a detailed discussion of the chaotic behavior in the Lorenz attractor, the Shil'nikov system, and the double scroll attractor.

3. Many new applications are included; previous applications have been updated.
4. There are now several chapters dealing with discrete dynamical systems.
5. We deal primarily with systems that are C^∞ , thereby simplifying many of the hypotheses of theorems.

The book consists of three main parts. The first part deals with linear systems of differential equations together with some first-order nonlinear equations. The second part of the book is the main part of the text: Here we concentrate on nonlinear systems, primarily two dimensional, as well as applications of these systems in a wide variety of fields. The third part deals with higher dimensional systems. Here we emphasize the types of chaotic behavior that do not occur in planar systems, as well as the principal means of studying such behavior, the reduction to a discrete dynamical system.

Writing a book for a diverse audience whose backgrounds vary greatly poses a significant challenge. We view this book as a text for a second course in differential equations that is aimed not only at mathematicians, but also at scientists and engineers who are seeking to develop sufficient mathematical skills to analyze the types of differential equations that arise in their disciplines. Many who come to this book will have strong backgrounds in linear algebra and real analysis, but others will have less exposure to these fields. To make this text accessible to both groups, we begin with a fairly gentle introduction to low-dimensional systems of differential equations. Much of this will be a review for readers with deeper backgrounds in differential equations, so we intersperse some new topics throughout the early part of the book for these readers.

For example, the first chapter deals with first-order equations. We begin this chapter with a discussion of linear differential equations and the logistic population model, topics that should be familiar to anyone who has a rudimentary acquaintance with differential equations. Beyond this review, we discuss the logistic model with harvesting, both constant and periodic. This allows us to introduce bifurcations at an early stage as well as to describe Poincaré maps and periodic solutions. These are topics that are not usually found in elementary differential equations courses, yet they are accessible to anyone with a background in multivariable calculus. Of course, readers with

a limited background may wish to skip these specialized topics at first and concentrate on the more elementary material.

Chapters 2 through 6 deal with linear systems of differential equations. Again we begin slowly, with Chapters 2 and 3 dealing only with planar systems of differential equations and two-dimensional linear algebra. Chapters 5 and 6 introduce higher dimensional linear systems; however, our emphasis remains on three- and four-dimensional systems rather than completely general n -dimensional systems, though many of the techniques we describe extend easily to higher dimensions.

The core of the book lies in the second part. Here we turn our attention to nonlinear systems. Unlike linear systems, nonlinear systems present some serious theoretical difficulties such as existence and uniqueness of solutions, dependence of solutions on initial conditions and parameters, and the like. Rather than plunge immediately into these difficult theoretical questions, which require a solid background in real analysis, we simply state the important results in Chapter 7 and present a collection of examples that illustrate what these theorems say (and do not say). Proofs of all of these results are included in the final chapter of the book.

In the first few chapters in the nonlinear part of the book, we introduce such important techniques as linearization near equilibria, nullcline analysis, stability properties, limit sets, and bifurcation theory. In the latter half of this part, we apply these ideas to a variety of systems that arise in biology, electrical engineering, mechanics, and other fields.

Many of the chapters conclude with a section called “Exploration.” These sections consist of a series of questions and numerical investigations dealing with a particular topic or application relevant to the preceding material. In each Exploration we give a brief introduction to the topic at hand and provide references for further reading about this subject. But we leave it to the reader to tackle the behavior of the resulting system using the material presented earlier. We often provide a series of introductory problems as well as hints as to how to proceed, but in many cases, a full analysis of the system could become a major research project. You will not find “answers in the back of the book” for these questions; in many cases nobody knows the complete answer. (Except, of course, you!)

The final part of the book is devoted to the complicated nonlinear behavior of higher dimensional systems known as chaotic behavior. We introduce these ideas via the famous Lorenz system of differential equations. As is often the case in dimensions three and higher, we reduce the problem of com-

prehending the complicated behavior of this differential equation to that of understanding the dynamics of a discrete dynamical system or iterated function. So we then take a detour into the world of discrete systems, discussing along the way how symbolic dynamics may be used to describe completely certain chaotic systems. We then return to nonlinear differential equations to apply these techniques to other chaotic systems, including those that arise when homoclinic orbits are present.

We maintain a website at math.bu.edu/hsd devoted to issues regarding this text. Look here for errata, suggestions, and other topics of interest to teachers and students of differential equations. We welcome any contributions from readers at this site.

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